
Role-Aware Dialogue Summarization for Podcasts Using Two-Stage Extraction and Generation

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Abstract

Podcasts are a representative form of long-form, multi-speaker dialogue, often containing information sharing, role-specific viewpoints, and implicit disagreements, which makes their automatic understanding and structured analysis challenging. In this project, we propose a role-aware podcast understanding pipeline that integrates summarization, viewpoint conflict detection and retrieval-based question answering under a weakly supervised setting.

For summarization, we adopt a two-stage approach consisting of a sentence-level extractor, and an abstractive refiner based on BART. The extractor identifies salient utterances from podcast transcripts, while the refiner generates coherent summaries from the selected content. The system produces a global summary as well as role-specific summaries for the host and the guest. Since the SPoRC dataset does not provide sentence-level importance annotations or human-written summaries, we rely on LLM-generated silver summaries and construct pseudo-labels using semantic similarity. This enables weakly supervision training on both stages.

Beyond summarization, we incorporate a viewpoint conflict detection module that applies a pretrained Natural Language Inference (NLI) model to host and guest summaries to identify potential stance disagreements. Sentence-level similarity and NLI predictions are further used to extract localized evidence supporting detected conflicts. We also demonstrate a lightweight retrieval-based question answering interface built on top of the generated summaries and conflict evidence.

Experiments on the SPoRC dataset show that the proposed approach achieves reasonable ROUGE-L performance across global and role-specific summaries, and can effectively identify episodes with divergent viewpoints. Overall the results demonstrate the feasibility of structured, role-aware podcast understanding without explicit sentence-level annotations.

1 Introduction

Podcasts are a rapidly growing source of long-form spoken content, often featuring multiple speakers engaging in extended discussions. Unlike news articles, podcast transcripts exhibit frequent topic shifts, informal language and complex speaker interactions, making automatic summarization a particularly challenging task. Listeners are often interested in not only a global summary, but also in understanding how different speakers contribute distinct perspectives.

Recent work in dialogue summarization highlights the importance of topic modeling and role-awareness for producing structured and interpretable summaries. However, training extractive

components typically requires costly sentence-level annotations, which are unavailable for large podcast datasets.

In this project we explore a role-aware two-stage summarization approach for podcasts using the SPoRC dataset. Our system first selects salient utterances using a neural extractor trained with pseudo-labels derived from semantic similarity to silver reference summaries. These selected utterances are then summarized using an abstractive BART-based generator. We support global summaries as well as role-specific summaries for hosts and guests, and evaluate our approach using ROUGE-L.

2 Related Work

2.1 Dialogue Summarization

Dialogue summarization differs from document summarization due to a turn-taking structure, speaker interactions and long-range discourse dependencies. Zou et al. [2021a] propose a saliency-aware topic modeling framework for spoken dialogue summarization, demonstrating that identifying topic-relevant segments significantly improves summary quality. Their work motivates the use of structured selection mechanisms prior to generation, which inspires the extractive stage of our pipeline.

2.2 Role-Oriented Summarization

Role-aware dialogue summarization aims to distinguish the contributions of different speakers in multi-party conversations. Guan et al. [2024] introduce interaction-aware contrastive learning to better separate speaker roles and improve role-specific summaries. While our implementation does not employ contrastive objectives, we adopt role filtering during pseudo-label construction and summary generation to enable host- and guest-specific summaries.

2.3 Topic-Aware Summarization

Topic-aware contrastive learning has been explored to improve abstractive dialogue summarization by encouraging topic consistency Liu et al. [2021]. Zou et al. [2021b] address low-resource dialogue summarization using multi-source pretraining, highlighting strategies for learning under limited supervision. Our work aligns with this direction by avoiding manual extractive annotations and instead relying on pseudo-labels derived from semantic similarity.

2.4 Code and Dataset Foundations

Our implementation is inspired by the open-source repository released by Zou et al.¹, which demonstrates extractive–abstractive pipelines for dialogue summarization. We also draw on the MediaSum dataset² as a reference for large-scale conversational summarization, though our experiments focus on SPoRC.

3 Approach

We propose a role-aware podcast understanding pipeline that integrates summarization, viewpoint analysis, and retrieval-based querying in a unified framework. The system is designed to operate under weak supervision, as no human-written summaries or sentence-level annotations are available in the SPoRC dataset.

3.1 Two-stage Summarization Framework

The summarization component follows a two-stage design consisting of a sentence-level extractor and an abstractive refiner. This separation allows the model to decouple content selection from surface-level generation.

¹<https://github.com/RowitZou/topic-dialog-summ>

²<https://github.com/zcgzcgzcg1/MediaSum>

The extractor operates at the utterance level and predicts the relative importance of each utterance within an episode. Utterances are encoded independently using a pretrained sequence-to-sequence encoder, and a lightweight classifier assigns an importance score to each utterance. Since no sentence-level supervision is available, the extractor is trained under weak supervision using pseudo-labels derived from episode-level summaries.

The refiner takes the utterances selected by the extractor and generates a coherent abstractive summary. Selected utterances are concatenated in their original temporal order and passed to a pretrained abstractive summarization model. To support multiple summarization objectives within a single architecture, control tokens are prepended to the input sequence to distinguish between global summaries and role-specific summaries for the host and the guest.

3.2 Pseudo-Label Construction

To enable training without manual annotations, we construct pseudo-labels using silver summaries generated by locally deployed large language models. Specifically, we use Qwen2.5 models to generate global, host-specific, and guest-specific summaries for each episode through task-specific prompts. These summaries are generated offline and serve as proxy supervision signals.

The silver summaries are used in two ways. First, they provide weak supervision for training the extractor by identifying utterances that are semantically similar to the summary. Second, they serve as reference targets for training the abstractive refiner. Although the summaries are model-generated, they provide a scalable and consistent supervision signal in the absence of human annotations.

3.3 Viewpoint Conflict Detection

Beyond summarization, the pipeline includes a viewpoint analysis module to identify potential stance disagreements between the host and the guest. Role-specific summaries are compared using a pretrained natural language inference (NLI) model, which predicts whether their semantic relationship corresponds to entailment, neutrality, or contradiction. Episodes predicted as contradictions are treated as containing potential viewpoint conflicts.

To improve interpretability, sentence-level conflict evidence is extracted by applying the same NLI model to semantically similar sentence pairs across host and guest summaries. These sentence pairs provide localized evidence supporting the global conflict prediction.

3.4 Retrieval-Augmented Question Answering

Finally, we design a lightweight retrieval-augmented question answering (QA) interface on top of the generated summaries and conflict evidence. Instead of relying on additional generative models, the QA module retrieves relevant pre-generated content based on semantic similarity between the user query and indexed summaries or evidence. This design enables interactive querying while maintaining controllability and interpretability.

4 Experiments

We evaluate summary quality using ROUGE-L F1 score, comparing generated summaries against silver reference summaries. Evaluation is performed separately for global, host, and guest summaries. We instantiate the proposed pipeline on the SPoRC dataset and evaluate it under a weakly supervised setting using LLM-generated silver summaries.

4.1 Dataset and Splits

Experiments are conducted on 1200 podcast episodes for which silver summaries are available. The data are split at the episode level into 1000 training episodes and 200 test episodes, and this split is fixed across all experiments. For each episode, three types of silver summaries are used: a global summary, a host summary, and a guest summary.

4.2 Experimental Configuration

The sentence-level extractor is implemented using the encoder of BART-large. Each utterance is truncated to a maximum length of 64 tokens. Pseudo-labels are constructed by selecting the top five utterances with the highest semantic similarity to the corresponding silver summary during training.

The abstractive refiner is also based on BART-large. Utterances selected by the extractor are concatenated and truncated to a maximum input length of 256 tokens. The maximum length of the generated summary is set to 150 tokens. Control tokens are prepended to the input to distinguish between global, host-specific, and guest-specific summaries.

Viewpoint conflict detection is performed using a pretrained DeBERTa-v3-large MNL model. Sentence embeddings for conflict evidence extraction and retrieval-based QA are computed using a pretrained sentence embedding model.

4.3 Training Procedure

Training is performed sequentially. The extractor is trained first using binary cross-entropy loss with logits, while keeping the pretrained encoder frozen and optimizing only the classification layer. After extractor training, the refiner is trained using the extractor-selected utterances as input and the silver summaries as targets, optimized with token-level cross-entropy loss. A single refiner model is shared across all summary types.

Due to the length of podcast transcripts and the computational cost of large pretrained models, training is conducted without episode-level batching.

4.4 Evaluation Protocol and Metrics

Summarization quality is evaluated using ROUGE-L by comparing generated summaries against the corresponding silver reference summaries. ROUGE-L is chosen for its tolerance to paraphrasing and word reordering, which is particularly suitable under weak supervision. As the references are model-generated, ROUGE-L scores are interpreted as approximate similarity measures rather than absolute indicators of summarization quality.

The extractor is not evaluated directly due to the absence of gold sentence-level annotations and is instead assessed indirectly through downstream summarization performance. In addition to intrinsic evaluation, qualitative analysis is performed on viewpoint conflict detection and retrieval-based QA results to assess the usefulness of the generated summaries in downstream tasks.

5 Results & Analysis

```
==== Train Extractor ====
[Extractor] global average loss: 0.6707
[Extractor] host average loss: 0.6515
[Extractor] guest average loss: 0.7337

==== Train Refiner (mixed) ====
[Refiner] global average loss: 2.6544
[Refiner] host average loss: 2.8052
[Refiner] guest average loss: 3.4431

==== Evaluate Refiner (ROUGE-L) ====
ROUGE-L (global): 0.2327
ROUGE-L (host): 0.2299
ROUGE-L (guest): 0.2023
```

Figure 1: Results on the test set

5.1 Summarization Performance

The summarization performance of the abstractive refiner is evaluated on the test set using ROUGE-L. As shown in Figure 2, the model achieves a ROUGE-L score of 0.2327 for global summaries, 0.2299 for host-specific summaries, and 0.2023 for guest-specific summaries.

Among the three summary types, global summaries obtain the highest ROUGE-L score. This result indicates that the model is able to capture the main semantic content of podcast episodes when summarizing across all speaker turns. The lower scores observed for role-specific summaries reflect the increased difficulty of summarizing content constrained to a single speaker role.

5.2 Comparison Between Host and Guest Summaries

Host summaries consistently outperform guest summaries in terms of ROUGE-L (0.2299 vs. 0.2023). This difference aligns with the conversational dynamics of podcasts, where hosts typically structure the discussion, introduce topics, and provide recurring contextual framing. As a result, host utterances tend to be more cohesive and easier to summarize.

Guest utterances, in contrast, often consist of detailed explanations, personal anecdotes, or episodic contributions. These characteristics introduce greater lexical and topical variability, which makes guest summaries more challenging to generate and evaluate using surface-level similarity metrics such as ROUGE-L.

5.3 Training Behavior Analysis

The observed performance differences are consistent with the training dynamics of both the extractor and the refiner. As shown in Figure 2, during extractor training, the average loss for global summaries is 0.6707, compared to 0.6515 for host summaries and 0.7337 for guest summaries. The higher loss for guest summaries suggests that the pseudo-label supervision is noisier for guest utterances, making importance prediction more difficult.

A similar pattern is observed during refiner training. The refiner achieves an average loss of 2.6544 for global summaries, while host and guest summaries exhibit higher losses of 2.8052 and 3.4431, respectively. These results indicate that generating coherent summaries from guest-only content is more challenging than summarizing global or host-focused content.

```
CONTRADICTION count: 2
===== PICKED EPISODE =====
Episode: https://anchor.fm/s/10858c10/podcast/play/15226445/https%3A%2F%2F3ctx1q1ktw2n1.cloudfront.net%2Fproduction%2F2020-5-15%2F82499051-44100-2-e7312cb0e01
Conflict label/score: CONTRADICTION 0.8907063007354736

Top evidence pairs: 1
- Host: Bob Gilligan introduces Pete Newburn, an Ecumenical officer with the Diocese of Joliet, as a speaker at a virtual prayer rally for racial justice.
- Guest: The host, Bishop Tate, leads the discussion, emphasizing the importance of recognizing and addressing systemic racism and the need for systemic change

===== QA =====
Q1: Bob Gilligan introduces Pete Newburn, an Ecumenical officer with the Diocese of Joliet, as a speaker at a virtual prayer rally for racial justice. He discu
Q2: Bob Gilligan introduces Pete Newburn, an Ecumenical officer with the Diocese of Joliet, as a speaker at a virtual prayer rally for racial justice. He discu
Q3: The podcast episode discusses the significance of COVID-19 and 1619, focusing on the historical context of these events and their impact on contemporary is
Q4: Yes. A disagreement between the host and the guest was detected.
Q5: The disagreement is reflected in the following statements:
- Host: Bob Gilligan introduces Pete Newburn, an Ecumenical officer with the Diocese of Joliet, as a speaker at a virtual prayer rally for racial justice.
- Guest: The host, Bishop Tate, leads the discussion, emphasizing the importance of recognizing and addressing systemic racism and the need for systemic change
```

Figure 2: QA & conffliction

5.4 QA & conffliction Analysis

In this episode, the conflict detection module predicts a CONTRADICTION label with a confidence score of 0.89, indicating a strong stance divergence between the host and the guest. The extracted evidence shows that the host focuses on introducing the speaker and contextualizing the event, while the guest emphasizes broader issues of systemic racism and social change. Although the disagreement is implicit rather than an explicit rebuttal, the contrasting communicative intents are successfully

captured by the model. The QA component correctly answers both content-related and conflict-related questions by retrieving relevant summaries and evidence. This example demonstrates that the proposed pipeline can detect viewpoint divergence, provide interpretable evidence, and support structured question answering over long-form podcast content.

6 Future Works & Limitations

While the proposed pipeline demonstrates the feasibility of role-aware podcast summarization and downstream analysis under weak supervision, several limitations remain and point to promising directions for future work.

First, the summarization models are trained and evaluated using LLM-generated silver summaries rather than human-written references. Although this choice is necessary due to the absence of gold summaries in the SPoRC dataset, it introduces potential bias, as the supervision and evaluation signals originate from the same source. Future work could mitigate this issue by adopting cross-model silver supervision, such as training and evaluating against summaries generated by different models or prompts, or by incorporating limited human evaluation to provide an additional independent assessment.

Second, the extractor component is trained using pseudo-labels derived from semantic similarity, which may be noisy, especially for guest utterances that exhibit higher topical diversity. This noise is reflected in the higher training loss and lower ROUGE-L scores observed for guest summaries. More robust pseudo-label construction strategies, such as adaptive selection thresholds or multi-reference supervision, could improve extractor reliability.

Third, the current conflict detection module relies on an off-the-shelf natural language inference model applied to generated summaries and sentence pairs. While effective in identifying many disagreement cases, it may conflate subtle semantic differences with true stance conflicts. Future work could explore conflict detection models explicitly trained on conversational disagreement data or incorporate discourse-level signals to improve precision.

Finally, the retrieval-based QA component is intentionally designed as a non-generative interface that retrieves pre-generated summaries and evidence. While this design ensures controllability and interpretability, it limits the expressiveness of answers. Future extensions could integrate lightweight generative models or hybrid retrieval-generation mechanisms to support more flexible and conversational question answering, while retaining structured grounding.

7 Conclusion

In this project we presented a role-aware pipeline for podcast understanding that integrates two-stage summarization and viewpoint conflict analysis. By combining pseudo-label supervised extractor with a BART-based abstractive refiner, the system is capable of producing global and role-specific summaries for hosts and guests. Additionally our conflict detection module leverages natural language inference to identify potential disagreements between speakers, providing interpretable sentence-level evidence.

Experimental results on the SPoRC dataset demonstrates that our approach achieves *reasonable* ROUGE-L performance across global and role-specific summaries, and successfully identifies episodes with conflicting viewpoints. The analysis highlights the challenges of summarizing guest contributions, emphasizing the impact of pseudo-label noise and the variability of speaker content.

Overall the project shows the feasibility of structured role-aware dialogue summarization under weak supervision, and provides a flexible foundation for downstream tasks such as retrieval based question answering. Future work can further improve the performance of our model by incorporating cross-model supervision, domain-adaptive extractors and advanced conflict detection techniques. This would enable richer and more reliable podcast understanding.

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